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HOW TO THINK ABOUT THE DIFFERENCES BETWEEN FROS AND PRIVATE ARPA PROGRAMS

Inside baseball differences between two institutional structures that seem similar from the outside

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It's easy to see the difference between ARPA-style programs and Focused Research Organizations as mere semantics. Each structure seeks to achieve similar goals — enabling public-good technical work that would otherwise languish; usually because it is too research-y or creates value that is too hard to capture for a startup, but requires more engineering effort, collaboration, or scale than academia can handle (more awesome sci-fi shit). Each structure is focus agnostic — capable of going after everything from brain-mapping to next generation electronics to climate change. Indeed, the line between them is nebulous: you could come up with dozens of ideas that either structure could enable. Each structure leverages empowered leaders: the CEO of an FRO or the PM of an ARPA program. Despite these similarities, it's important to distinguish between them because the *how* matters: some ideas that would thrive as an ARPA program would wither as an FRO and vice versa. We could think of different ideas as being either more FRO shaped or more ARPA shaped (and of course there are many ideas that are neither FRO nor PARPA shaped — let's not start thinking that everything looks like a nail).

> It's important to note that *the comparable unit of research management is between a single FRO and a single ARPA-style program*. FROs themselves are program or project-level organizations. I'll start by comparing FROs and ARPA-style programs directly, but it's worth noting that PARPA's hypothesis is that it's important to start and run multiple ARPA-style programs under a single organizational umbrella, while FROs have much more independence.

At a high level, FRO-shaped ideas are ready to focus on execution from day one. ARPA programs require more path discovery before switching to execution mode. This idea-level difference cashes out in several differences in execution.

idea of what success means and a strong hypothesis about what work will get you there. PARPA programs ramp more slowly, from ~$200k over ~six months to ~$5 million over ~18 months to $30m, FRO-level funding, with go/nogo gates between different stages. Most ARPA-style program do *aspire* to get to FRO levels of funding, but the nature of their activities (more initial uncertainty about goals, correct approaches, and feasibility) means that many won't get there and even those that will wouldn't effectively use that much money effectively from day one.

FROs and ARPA-style programs have different org charts. (Org charts matter!) FROs have a core team that does all of the research design and large chunk of the work, perhaps contracting some execution out. This core team is ideally autonomous from the FRO's funding source. In ARPA-style programs the "core team" is a single PM, who coordinates external collaborators doing different projects. These collaborators do all of the work and a good chunk of research design. Over time, an ARPA-style program could internalize work as coordination costs go up, but doing that is not a given. Like funding shapes and other differences, the different structures are meant to address different sorts of work. Distributed work enables ARPA-style programs to try several tracks in parallel without internal politics: this is important because ARPA-style programs involve much more uncertainty about approaches. The external work also allows an ARPA-style program to re-factor, re-combine, or prune different approaches or projects without anybody leaving their home org or being fired. External research can give ARPA-style programs access to expertise and equipment that would otherwise be hard to hire or purchase. FRO's internalized research enables tight iteration loops and complete focus on the project which are some of the most important things when success is primarily a matter of execution.

> A hard truth is that the differences in both funding and org structures of FROs and ARPA-style programs make it much more appealing to run an FRO, all things held equal. FROs give you both more direct control and a larger pool of money to play with sooner. For those of us who just want there to be more counterfactually impactful work, this incentive bias means that it's especially important to be honest about which things will make poor FROs but good PARPA programs.

To distill these differences into discriminating questions:

1. Can you describe every step of the process to achieve the goal in executable detail?
2. Is it clear which overall approach to tackle the idea is the right one?
3. Does the project need extremely rare equipment or expertise?
4. Does the project look like putting together a number of lego pieces into a working system?
5. Do you want to explore a number of possible high-level system concepts or general approaches before committing to a specific goal? The crux question of program design is when to commit — FROs need strong conviction that it's worthwhile to commit from day one.
6. Is part of the impact indirect? Will a chunk of the counterfactual impact come from kicking off a discipline, creating a community, or showing that something is possible?

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organizations: some of the researchers might technically be employees of the funding organization or be contracted from a more permanent research organization. At the same time, I might say "ARPA-style programs are distributed" but it might make sense to build a team to do a specific piece of the work.

- FROs
  - Clear goal from day one
  - Tight coupling between work-streams from day one
  - Strong conviction about the right approach at the project level early on
  - Most work done under one organizational roof, outsourcing routine aspects to contractors but not widely distributing the fundamental research questions.
  - Quickly ramp up to $10-50m range to focus on scalable execution not achievable in existing organizational settings.
- PARPA Programs
  - Might take 6 months to a year to nail down a goal
  - Projects are initially parallel with coupling increasing over time
  - Unclear about the right approach
  - Naturally leverages existing organizations and forms networks among them to develop new system level ideas, approaches or concepts.

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